CMOS Design Styles

Ranga Vemuri

Pass Transistor Logic

MOS Capacitances

Strong and Weak Signals

Signal Transmission Through n-Transistor

n-transistor

- Good conductor of logic “0”
- Poor conductor of logic “1”

Signal Transmission Through p-Transistor

p-transistor

- Good conductor of logic “1”
- Poor conductor of logic “0”

Pass Transistor Logic – Transmission Quality

Transmission Gate (T-gate)

Transmission Gate – Transmission Quality

Bidirectionality and Charge Sharing

Multiplexor

Simple Memory: Charge Sharing

Issues With T-Gate

- Transistors – doubled
- Needs inverter.
- Routing
- Charge sharing
- Lack of Noise Immunity

Fully Complemented Static CMOS Inverter

Fully Complemented Static CMOS NAND

Fully Complemented Static CMOS NOR

Fully Complemented Static CMOS Complex Gate

- Pull-up network (PUN) uses the p-transistors only.
- Pull-down network (PDN) uses the n-transistors only.
- PUN and PDN are logical complements of each other.

Structural Complements (to obtain logical complements)

To Derive PUN from PDN:

- Replace n-transistors with p-transistors.
- Replace series connection with parallel connection and vice-versa.
- Replace GND by VDD.

Minority of 3 in Static CMOS

Majority of 3 in Static CMOS

EX-OR in Static CMOS

How many levels of static logic are sufficient to implement any Boolean function?

Memory in Static CMOS

Disadvantages of Fully Complemented Static CMOS

- Doubling of number of transistors when compared with pass transistor logic.
- Vdd and GND to be routed to each gate.